

Julia Gaubatz Tao

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Technical Summary

Aerospace engineer and MIT Ph.D. candidate with experience in aircraft design, alternative-fuel aircraft safety, systems architecture, and flight vehicle testing. Developed methods for integrating hydrogen and methane fuel-system safety constraints into early-stage multidisciplinary aircraft design optimization. Hands-on experience in small UAV prototyping, wind-tunnel testing, mechanical design, manufacturing, and flight testing.

Education

Massachusetts Institute of Technology

Ph.D., Aeronautics and Astronautics

Cambridge, MA

Expected Sept. 2026

- Dissertation: *Integrating Safety Constraints into Early-Stage Design of Hydrogen- and Methane-Fueled Aircraft*.
- Advisor: R. John Hansman.

Massachusetts Institute of Technology

S.M., Aeronautics and Astronautics

Cambridge, MA

May 2023

- Thesis: *Design and Development of Stability and Control Systems for Small, Deployable Aircraft*.

Massachusetts Institute of Technology

S.B., Aerospace Engineering; Minor in Mathematics

Cambridge, MA

May 2021

- GPA: 4.8/5.0.

Tools & Methods

Software/Analysis: Python, MATLAB, SolidWorks, AVL, XFOIL, QPROP, QMIL.

Testing/Fabrication: Wind-tunnel testing, UAV flight testing, machining (lathe, mill, waterjet), composite layups, FDM/SLA 3D printing.

Research and Engineering Experience

MIT International Center for Air Transportation

Ph.D. Researcher, Hydrogen and Methane Aircraft Fuel-System Safety

Cambridge, MA

Aug. 2022–Present

- Developed methodology for incorporating fuel-system safety constraints into early-stage hydrogen and methane aircraft design.
- Conducted structured hazard identification and prioritization using the FAR 25.1309 safety framework, producing a fuel-system hazard list with estimated risk levels and mitigation strategies.
- Introduced the “designated leak zone” as a formal hazard-zone classification and certification concept for hydrogen and methane aircraft, analogous to the designated fire zone in current regulations.
- Applied fire, explosion, and cryo-mechanical hazard models to aircraft-scale hydrogen and methane fuel leaks, parameterized by design-relevant variables.
- Developed aircraft design model integrating passive leak-mitigation strategies and aircraft geometries, enabling transport aircraft optimization subject to leak safety constraints.
- Quantified the transport energy impact of passively mitigating leaks of differing sizes for hydrogen and methane aircraft across two fuel tank configurations.

MIT International Center for Air Transportation

S.M. Researcher, Small UAV Stability and Control System Design

Cambridge, MA

June 2021–May 2023

- Designed the stability and control system for Firefly, a deployable micro-UAV powered by a slow-burning solid rocket motor, utilizing a pull-spring-actuated folding tail with oblique hinge serving as both deployment and actuation axis.
- Developed multiple 3D-printable vehicle iterations to rapidly evaluate folding-tail configurations.
- Built pitch-balance test setup and evaluated longitudinal stability and controllability in wind-tunnel testing.
- Conducted unpowered drop tests from quadcopter to collect flight data.
- Compared wind-tunnel and drop test results against AVL vortex-lattice predictions.

MIT Lincoln Laboratory, Group 77 – Rapid Prototyping

Summer Research Program Intern

Lexington, MA

June 2023–Aug. 2023

- Conducted preliminary design of a 60-ft wingspan UAV for carriage of a large sensor payload under demanding functional and trajectory requirements.
- Evaluated unconventional UAV configurations against payload integration, aerodynamic performance, structural feasibility, and mission constraints.

SpaceX

Hawthorne, CA

Turbomachinery Intern, Raptor Oxygen Pump Hardware Design

June 2020–Aug. 2020

- Conducted preliminary mechanical design of oxygen turbopump inducer for Raptor engine revision, addressing axial and torque loads during operation and assembly while improving manufacturability and serviceability.
- Investigated electrostatic discharge damage in rotor group and assessed ESD mitigation performance.

Blue Origin

Kent, WA

Turbomachinery Intern, BE-4 Seal Test Rig Design

June 2019–Aug. 2019

- Analyzed dynamic-seal behavior in the BE-4 main turbopump using engine test data.
- Designed test rig and produced CAD models and manufacturing drawings to simulate operational environment and flow behavior.
- Developed test campaign and communicated plan to turbomachinery design and engine test teams.

Selected Engineering Projects

Solar Electric Seaplane

Spring 2023

Teaching Assistant, MIT 16.821: Flight Vehicle Development

- Supported student teams through component design, fabrication, integration, and flight testing of a 4-meter wingspan solar-electric seaplane.
- Taught laboratory manufacturing techniques, including composite layups.
- Coordinated assignments, grading, and communication between students and faculty.

Desktop Lathe

Spring 2022

Student, MIT 2.720: Elements of Mechanical Design

- Designed, machined, and tested a desktop lathe from functional requirements through validation.
- Used homogeneous transformation matrices to model stiffness and derive component requirements.
- Demonstrated cutting of aluminum part with 0.001 inch diametral accuracy and 0.0009 inch repeatability.

Solar HALE Aircraft Propulsion

Sept. 2020–May 2021

Student, MIT 16.82/821: Flight Vehicle Engineering

- Configured propulsion system for “Dawn 1,” an ozone-monitoring high-altitude, long-endurance aircraft.
- Developed MATLAB tools to design and analyze propulsion configurations.
- Fabricated propeller section layups to evaluate structural performance and manufacturability.

Rocketry – Level 3 Certification and MIT Rocket Team Leadership

Sept. 2017–Aug. 2020

President and Propulsion Lead, MIT Rocket Team

- Designed, scratch-built, launched, and recovered an M-class solid-motor rocket, earning Tripoli Level 3 certification; vehicle reached over 14,000 ft and 950 mph.
- Served as President of MIT Rocket Team, leading approximately 80 students in development of a P-class rocket.
- Coordinated design reviews, served as liaison to MIT administration and industry partners, and obtained FAA waivers for flights up to 80,000 ft.
- Served as Propulsion Lead from Jan. 2020–July 2020.

Selected Publications

- Tao, J. and Hansman, R. J., “Fuel Leak Hazard Management for Hydrogen and Methane Aircraft,” *Second US-Europe Air Transportation Research and Development Symposium*, Delft, Netherlands, 2026.
- Gaubatz, J., et al., “Estimating the Energy Demand of a Hydrogen-Based Long-Haul Air Transportation Network,” *2023 International Conference on Future Energy Solutions*, Vaasa, Finland, 2023. doi: 10.1109/FES57669.2023.10182543.

Honors

- **MIT Energy Initiative Energy Scholar**, 2025–2026; competitive fellowship for graduate research advancing the low- or zero-carbon energy transition.
- **Brooke Owens Fellow**, Class of 2019; nationally competitive aerospace fellowship, hosted by Blue Origin.